

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our advanced predictive analysis has demonstrated strong performance, accurately classifying over 83% of outcomes. This high level of reliability ensures that the insights derived from this model are robust and trustworthy, providing a solid foundation for data-driven strategic decisions. The analysis highlights three key factors significantly influencing the predicted outcomes: gender (specifically male), the price paid for a service or product, and age. Gender appears to be the most influential factor, followed closely by pricing and age. Given their substantial impact, we strongly recommend continuously monitoring these key drivers. Tracking changes in these variables will be crucial for anticipating shifts in outcomes, enabling proactive strategic adjustments, and optimizing our resource allocation to maximize business results.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

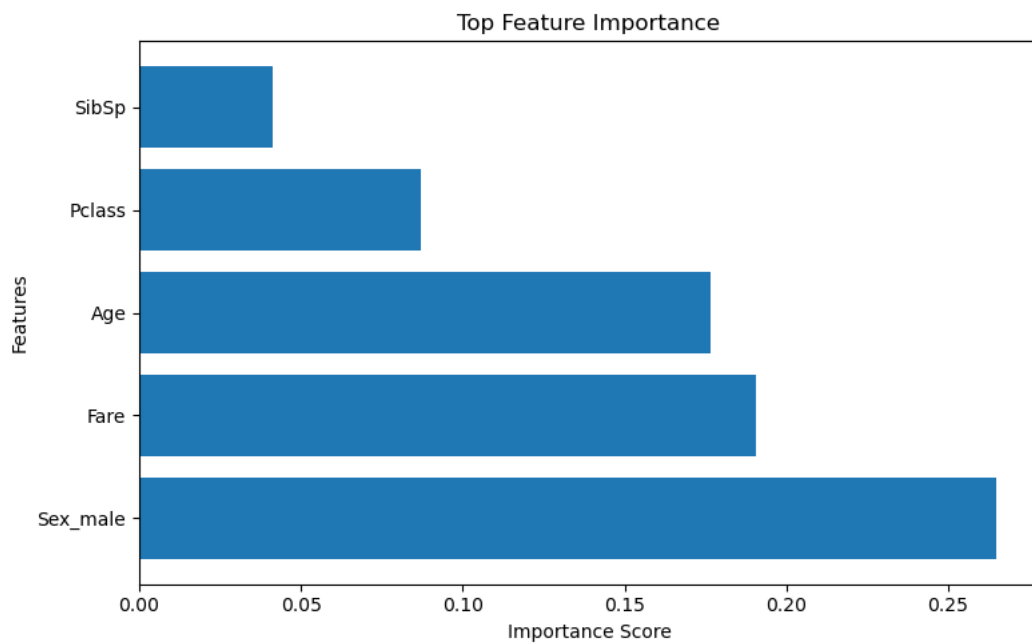
Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.

- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

